

UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

Object Oriented Programming

36863

ConixCoffeeAndIceCreamShop - Product Backlog

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Product Backlog

ID	Priority	User Story / Description	Source	Acceptance Criteria (Summary)
FR-01	Must	As the owner, I want to log in with a username and password so that the system data remains secure.	PB-01	Secure authentication; session protection.
FR-02	Must	As the owner, I want to add new products to the menu so that the digital catalog stays up to date.	PB-02	Product name, price, description, and status can be entered and saved.
FR-03	Must	As the owner, I want the system to calculate the estimated production cost of a product based on its ingredients and their costs.	PB-03	Cost is automatically computed from ingredient quantities and unit costs.
FR-04	Must	As the owner, I want the system to calculate the profit margin of each product based on production cost and selling price.	PB-04	Margin = (Selling price – Production cost) / Selling price, displayed clearly.
FR-05	Must	As the owner, I want to define the ingredients and quantities required to prepare each product.	PB-13	Recipe (bill of materials) can be associated with each product.
FR-06	Must	As the owner, I want to increase inventory quantities when new supplies are purchased.	PB-14	Stock levels update immediately after purchase registration.
FR-07	Must	As an employee, I want the system to automatically calculate the total cost of an order based on selected products and quantities from a digital menu.	PB-09	Order total updates in real time; final total is accurate.
FR-08	Must	As an employee, I want the system to automatically add the corresponding packaging cost when an order is marked as takeaway.	PB-10	Packaging cost is added only for takeaway orders.
FR-09	Must	As the system, it shall automatically deduct the ingredients and supplies used in an order from the available inventory.	PB-11	Inventory is reduced according to product recipes immediately after order confirmation.

FR-10	Must	As the system, it shall prevent or clearly indicate the sale of products whose required ingredients are unavailable or insufficient.	PB-15	Out-of-stock products are disabled or flagged before order submission.
FR-11	Must	As the owner, I want the system to notify me when an ingredient or inventory item reaches its minimum stock level.	PB-12	Alert is shown promptly when stock ≤ minimum threshold.
FR-12	Must	As the owner, I want to record transactions made through electronic banking and in cash.	PB-05, PB-06	Both payment methods can be registered and linked to orders or standalone income.
FR-13	Must	As the owner, I want to register business expenses, including purchases of raw materials and other supplies.	PB-08	Expense category, amount, date, and description can be recorded.
FR-14	Must	As the owner, I want the system to maintain a chronological record of the business's income and expenses.	PB-07	All income and expense entries are listed in date order and can be filtered/reviewed.
NFR-01	Must	The system shall provide an intuitive and easy-to-use interface that allows employees to create orders with minimal training.	NFR-01	Clear navigation, digital menu, and simple order flow.
NFR-02	Must	The system shall calculate order totals and update inventory within a maximum of 2 seconds after an order is submitted.	NFR-02	Response time ≤ 2 seconds under normal load.
NFR-03	Must	The system shall prevent the loss of transaction, inventory, and product information in the event of an unexpected system failure.	NFR-03	Data persistence and recovery mechanisms in place.
NFR-04	Should	The system shall maintain records of relevant changes to products, inventory, expenses, and transactions for later review (audit trail).	NFR-04	Change history is stored and accessible.
NFR-05	Must	Notifications and inventory updates shall be generated promptly after a transaction is completed.	NFR-05	Updates and alerts occur immediately after confirmation.